Electric Drives and Machines Institute
Graz University of Technology

Laboratory course

Control of Electric Drives and Machines

Course-Nr. 431.312

# Unit 3:

# Parameter of Induction Machine

**Teaching goals**

Parametrise the model of the induction machine.

**In the lab**

Apply standard test procedures to determine the parameters of the equivalent circuit diagram and indentify converter voltage drop.

# 1 Steady state

Constant mechanical speed and

$$\underline{v}_S^S = \widehat{v}_S e^{j\varphi_{vS}} e^{j\omega_S t} = \widehat{\underline{V}}_S e^{j\omega_S t} \tag{1}$$

$$\begin{aligned} \widehat{\underline{V}}_S &= R_S \widehat{\underline{I}}_S + j\omega_S \left( L_{\sigma S} \widehat{\underline{I}}_S + \widehat{\underline{\Lambda}}_m \right) \\ 0 &= R'_R \widehat{\underline{I}}'_R + j\omega_R \left( L'_{\sigma R} \widehat{\underline{I}}'_R + \widehat{\underline{\Lambda}}_m \right) \end{aligned} \tag{2}$$

With $\omega_R = \omega_S - p \cdot \omega_{mech}$.
Rotor voltage equation is divided by the slip $s$

$$s = \frac{\omega_R}{\omega_S} \tag{3}$$

Substituting magnetising flux linkage by $L_m \left( \widehat{\underline{I}}_S + \widehat{\underline{I}}'_R \right)$ gives

$$\begin{aligned} \widehat{\underline{V}}_S &= R_S \widehat{\underline{I}}_S + j\omega_S L_{\sigma S} \widehat{\underline{I}}_S + j\omega_S L_m \left( \widehat{\underline{I}}_S + \widehat{\underline{I}}'_R \right) \\ \frac{\widehat{\underline{V}}'_R}{s} &= \frac{R'_R}{s} \widehat{\underline{I}}'_R + j\omega_S L'_{\sigma R} \widehat{\underline{I}}'_R + j\omega_S L_m \left( \widehat{\underline{I}}_S + \widehat{\underline{I}}'_R \right) \end{aligned} \tag{4}$$

The equations (4) can be represented by an equivalent circuit diagram.

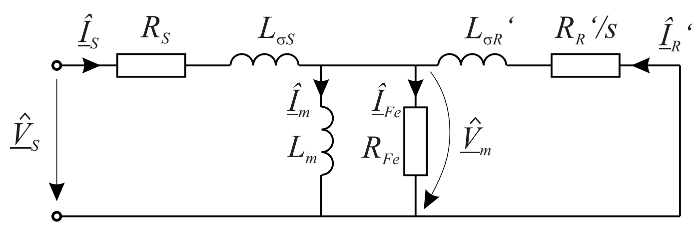


Figure 1: Equivalent circuit diagram for the induction machine with squirrel cage rotor

# 2 Tests

## 2.1 Nameplate data

| Name | Value |
|---|---:|
| Voltage | 124 V |
| Current | 21.4 A |
| Power | 3.3 kW |
| Frequency | 150 Hz |
| Speed | 4265 rpm |

Table 1: Nameplate data for star connection

## 2.2 Locked rotor test

As the name suggests, rotor should be locked. But to take care of the commutator of the DC-machine, reference speed of the speed controlled DC-machine is set to a small negative value, e.g. -10 rpm. A rotating stator voltage space vector is applied to the induction machine. The goal should be to measure rotor behaviour at frequencies that occur during normal operation, i.e. a few Hz. But stator frequencies normally are set to higher values to ensure that inductances determine the partition of stator current into rotor- and magnetising current. To calculate the impedance for the fundamental harmonic, a fourier analysis is applied to the measured phase voltages and currents. Averaging over the three phases is applied (this helps to diminish the influence of negative sequence system).

$$\begin{aligned} \widehat{V}_S &= \frac{\widehat{V}_a + \widehat{V}_b + \widehat{V}_c}{3} \\ \widehat{I}_S &= \frac{\widehat{I}_a + \widehat{I}_b + \widehat{I}_c}{3} \end{aligned} \tag{5}$$

With stator power $P_S = P_a + P_b + P_c$ according to fundamentals phase shift $\varphi_{vi}$ can be calulated

$$\cos(\varphi_{vi}) = \frac{2 \cdot P_S}{3 \cdot \widehat{V}_S \cdot \widehat{I}_S} \tag{6}$$

and locked rotor impedance $\underline{Z}_{LR} = R_{LR} + j \cdot \omega_S \cdot L_{LR}$ is determined by

$$\underline{Z}_{LR} = \frac{\widehat{V}_S}{\widehat{I}_S} \cdot e^{j \cdot \varphi_{vi}} \tag{7}$$

For a simplified evaluation the magnetising current can be neglected.

$$\underline{Z}_{LR} = \left( R_S + \frac{R_R'}{s} \right) + j \cdot \omega_S \left( L_{\sigma S} + L_{\sigma_R}' \right) \tag{8}$$

Without additional knowledge it is not possible to split up the inductance $L_{LR}$ into stator- and rotor leakage inductance.

Without referring the rotor current to stator winding flux equations are given by

$$\begin{aligned} \underline{\lambda}_S^S &= L_s \cdot \underline{i}_S^S + M \cdot \underline{i}_R^S \\ \underline{\lambda}_R^S &= M \cdot \underline{i}_S^S + L_R \cdot \underline{i}_R^S \end{aligned} \tag{9}$$

Now an arbitrary chosen transfer-ratio $k$ is introduced

$$\underline{i}_R^S = k \cdot \underline{i}_R^{S'} \tag{10}$$

- $L_m = k \cdot M \Rightarrow k = \frac{L_m}{M}$
  This leads to the T-equivalent circuit

$$\begin{aligned} \underline{\lambda}_S^S &= L_s \cdot \underline{i}_S^S + L_m \cdot \underline{i}_R^{S'} \\ \underline{\lambda}_R^{S'} &= L_m \cdot \underline{i}_S^S + L_R' \cdot \underline{i}_R^{S'} \end{aligned} \tag{11}$$

- $L_S = k \cdot M \Rightarrow k = \frac{L_S}{M}$
  This leads to the Γ-equivalent circuit

$$\begin{aligned} \underline{\lambda}_S^S &= L_s \cdot \underline{i}_S^S + L_S \cdot \underline{i}_R^{S'} \\ \underline{\lambda}_R^{S'} &= L_S \cdot \underline{i}_S^S + L_R' \cdot \underline{i}_R^{S'} \end{aligned} \tag{12}$$

- $M = k \cdot L_R \Rightarrow k = \frac{M}{L_R}$
  This leads to the inverse Γ-equivalent circuit

$$\begin{aligned} \underline{\lambda}_S^S &= L_s \cdot \underline{i}_S^S + \frac{M^2}{L_R} \cdot \underline{i}_R^{S'} \\ \underline{\lambda}_R^{S'} &= \frac{M^2}{L_R} \cdot \underline{i}_S^S + \frac{M^2}{L_R} \cdot \underline{i}_R^{S'} \end{aligned} \tag{13}$$

All these equivalent circuits are equivalent! If the slot geometries are standard ones, the ratio $L_{\sigma S}/L'_{\sigma R}$ is in the range from 0.66 to 1.5 and a ratio of $L_{\sigma S}/L'_{\sigma R} = 1$ may be assumed.

With the parameters $R_{LR}$ and $L_{LR}$ from locked rotor test a rough PI stator current controller can be designed to facilitate the further experiments.

## 2.3 Stator resistance

Stator resistance is identified by applying constant stator current space vectors, e.g. at an angle of 90°, 210° and 330°, and measuring the average values of phase voltages and currents.
Assumption: No skin effect (wire wound coils).

## 2.4 Noload test

A constantly rotating voltage space vector is applied. If additional to the magnetising curve frictional torque is of interest too, the noload test in motor operation is preferred. If not, then the induction machine is operated as a generator. That means, that another machine has to develop the torque to overcome frictional losses. In each case the noload impedance is measured at constant stator frequency but different stator currents. Measurements of phase voltages and phase currents have to be evaluated to calculate the fundamental harmonics similar to the procedure applied for locked rotor test.

- Generator operation
  Since the rotor runs at synchronous speed, the fundamental wave of airgap flux density is at standstill relative to the rotor. No rotor current flows, no torque is developed. But stator iron is exposed to a rotating flux density distribution and losses will arise. For their consideration an iron loss resistance parallel to magnetising inductance is included in the equivalent circuit.

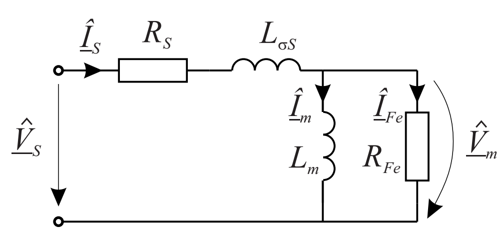


Figure 2: Equivalent circuit diagram for the noload test in generator operation

- Motor operation
  Again noload impedance is to be determined for the fundamental frequency. Indcution machine has to overcome frictional torque by itself but the rotor is close to synchronous speed. Due to this the rotor current is almost in phase to the voltage drop over the magnetising inductance. Stator current can be split up into one part that is in phase with magnetising voltage $\widehat{\underline{V}}_m$ and one that is shifted by 90° to it, this is the magnetising current $\widehat{\underline{I}}_m$.

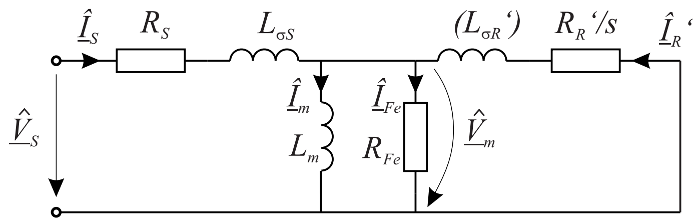


Figure 3: Equivalent circuit diagram for the noload test in motor operation

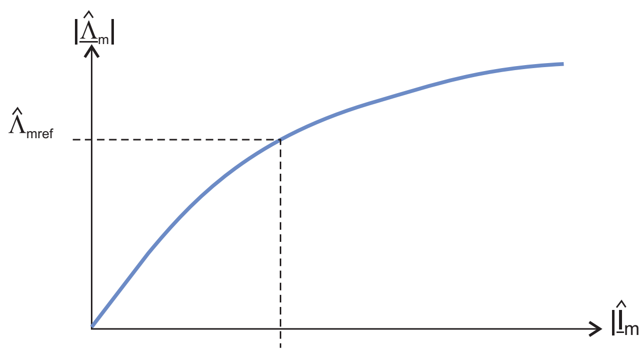


Figure 4: Magnetising curve

The outcome is the magnetising curve. Magnetising inductance should be determined at a magnetising fluxlinkage of $\widehat{\underline{\Lambda}}_{mref} = 0.08Vs$.

## 2.5 Converter voltage drop

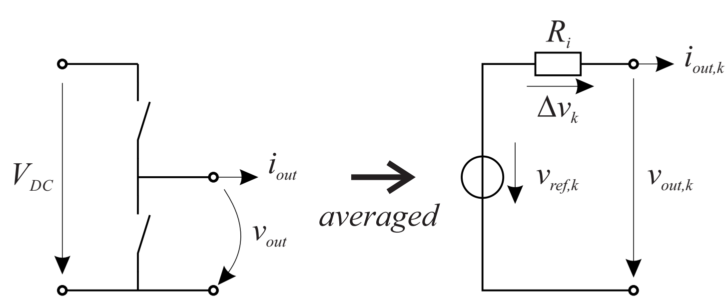


Figure 5: Equivalent representation of half bridge.

Quantities averaged over the k-th pulse period are given by

$$\begin{aligned} v_{out,k} &= \frac{1}{T_{sw}} \int_{T_{SW}} v_{out} dt \\ \Delta v_k &= \frac{1}{T_{sw}} \int_{T_{SW}} R_i\left(i_{out}\right) \cdot i_{out}\left(t\right) dt \end{aligned} \tag{14}$$

The reference voltage is given by

$$v_{ref,k} = V_{DC} \cdot d_k \tag{15}$$

with duty cycle $d_k$. The output current $i_{out,k}$ is the current sampled at the time instant $t_k = T_{SW} \cdot k$ with $k = 0,\ 1,\ 2 \ldots.$

Similar to the experiment done for the DC-machine converter, stator current space vectors are applied to keep one of the phase currents at zero, e.g. in direction 30° as shown below (but we will apply 90°, 210° and 330°).

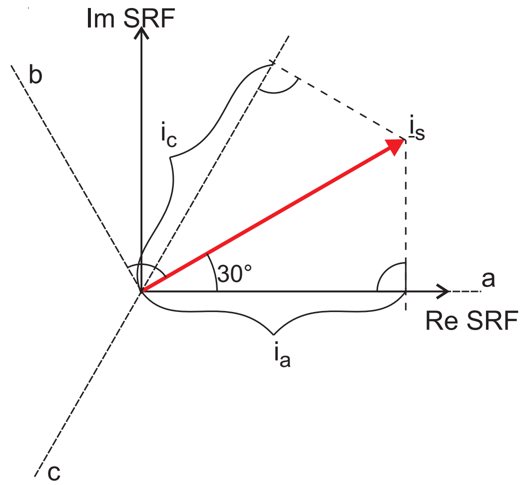


Figure 6: Stator current space vector to keep phase current $i_b$ at zero.

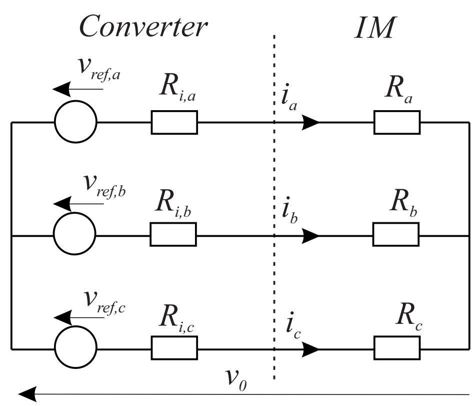


Figure 7: Equivalent circuit for converter voltage drop measurement.

### 2.6 Test circuit

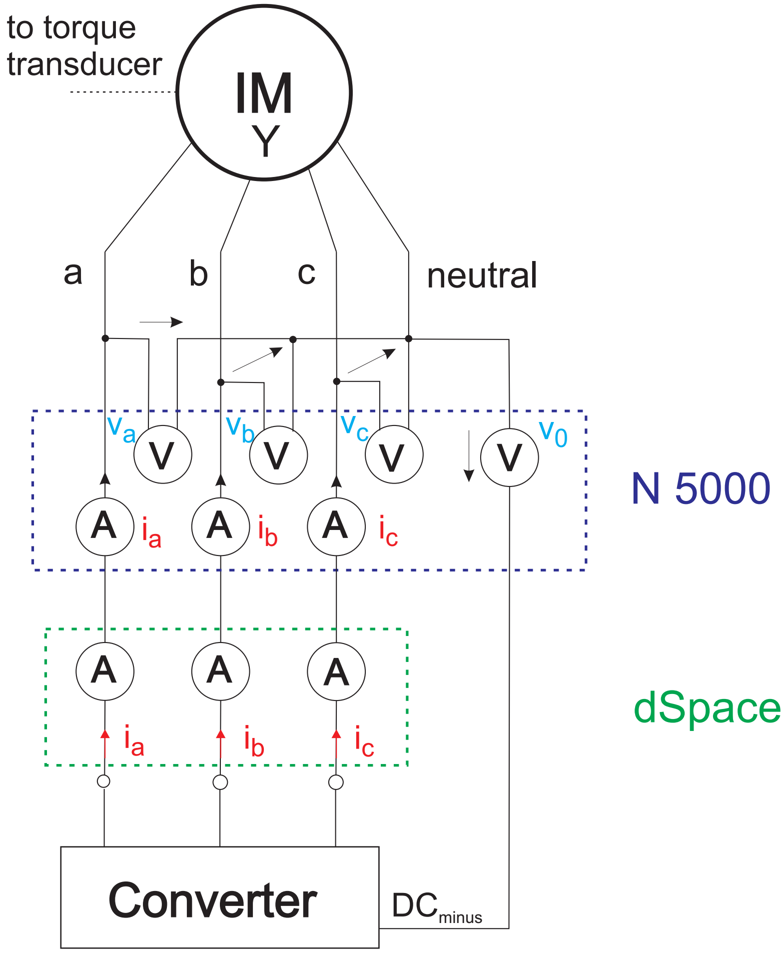


Figure 8: Test circuit